Example: Modified problem of a strong explosion.—An example of a self-similar intermediate asymptotics is provided by the phenomena occurring at the initial stage of a nuclear explosion.

At the very first stage, immediately following energy release, thermal waves are propagated in as yet stationary gas; heat conduction is then determined mainly by radiation.[2]

The rate of transfer of energy by radiation decreases with time and ultimately becomes small in comparison with the speed of sound; then there comes the next stage where it is necessary to consider the motion of the gas, which may be assumed to be adiabatic. In the heated gas, the pressure increases greatly with the result that a strong shock wave is generated and propagated through the gas (the pressure at the shock front being several orders higher than the initial gas pressure). When the shock wave overtakes the thermal wave (their common radius is taken to be R_0 at that instant), the hydrodynamic stage of the explosion sets in. At this stage, the solution also depends on the total amount of released energy E, the initial gas pressure p_0, and its initial density ρ_0; it has the following parameters with dimensions of length: $L_1=(E/p_0)^{1/3}$, $L_2=R_0$, and with dimensions of time: $t_1=(\rho_0 E^{2/3}/p_0^{5/3})^{1/2}$, $t_2=(\rho_0 R_0^5/E)^{1/2}$. For the conditions of a nuclear explosion, $L_1 \gg L_2$ and $t_1 \gg t_2$. The intermediate asymptotic regime corresponding to the stage of strong shock waves was discovered and studied by Sedov (1946) and

[2] Transfer of energy by radiation at the initial stage occurs at a rate many times the speed of sound so that the hydrodynamic displacement of substance at this stage can be neglected.